Kien Le

Tallahassee, FL | lekien140405@gmail.com
github.com/kienle1404 | linkedin.com/in/kienle414 | kienle1404.github.io

Summary

Graduate researcher at Florida State University working on trustworthy AI: model extraction and ownership verification for LLMs and GNNs, privacy-preserving federated learning for healthcare, and compiler-level systems optimization. I aim to build machine learning systems that are secure, private, and efficient.

Education

2026 – 2028 **Florida State University**, M.S. in Computer Science (Expected 05/2028)

2023 – 2026 **Florida State University (Honors)**, B.S. in Computer Science (GPA: 3.875/4.0)
Relevant Coursework: Data Structures & Algorithms, Probability, Operating Systems, Data Mining.
Honors: President's List (S24, S25), Dean's List (F23, F24, F25)

Research Experience

Research Assistant, Triple-I Lab 01/2026 – Present

- Advised by Dr. Tuy Nguyen; built a personalized federated ECG classifier in PyTorch: a dual-attention transformer adaptor on a frozen ECG foundation model (ECGFounder), reaching 85.8% per-client Micro-F1 on FedCVD, 13–26 points above personalization-free baselines incl. LoRA.
- Added CKKS homomorphic encryption on shared parameters only: 71.5% smaller payload at 1.06× overhead; gradient inversion failed to recover ECG waveforms (first author, *Healthcare Analytics* 2026; MWSCAS 2026).

Research Intern, Oak Ridge National Laboratory 05/2026 – 07/2026

- Mentored by Dr. Massimiliano Lupo Pasini and Dr. Haoran Niu; implemented a temporal graph convolutional network (T-GCN) for anomaly detection in FNET/GridEye power grid data.
- Integrated the model into HydraGNN, ORNL's distributed PyTorch GNN framework, and trained it across multiple GPUs to scale graph training.

Research Assistant, Compiler Optimization 07/2025 – 01/2026

- Advised by Dr. David Whalley; worked on backend optimizations in the Very Portable Optimizer (VPO), including copy propagation, branch chaining, and tail duplication.
- Analyzed how transformations affect performance and correctness across benchmarks.

Researcher, Large Language Model Security 03/2025 – 12/2025

- Advised by Dr. Yushun Dong; co-designed a query-efficient attack for extracting domain knowledge from black-box LLMs (AAAI 2026).
- With 500 queries, a LoRA-tuned LLaMA-3.1 8B surrogate rose from 49.0% to 62.0% on MedQA, beating two baselines.

Researcher, GNN Security 08/2024 – 04/2025

- Advised by Dr. Yushun Dong; evaluated query-based model extraction and ownership verification attacks on GNNs using PyTorch Geometric baselines (GraphSAGE, GIN).
- Studied robustness and ownership verification of graph learning systems in adversarial settings.
- Contributed an open-source implementation of **GNNFingers** to the **PyGIP library**, improving reproducibility for GNN security research.

Publications

[1] **Kien Le**, Quoc Bao Phan, Nhat Minh Nguyen, Tuy Tan Nguyen. "**An Encrypted Dual-Attention Federated Learning Framework for Personalized Electrocardiogram Classification**." *Healthcare*

Analytics, vol. 10, 100482, 2026. [Paper] [Code]

[2] **Kien Le**, Joseph Lindley, Quoc Bao Phan, Tuy Tan Nguyen. "**Dual Attention Heads for Personalized Federated Learning in ECG Classification**." *IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2026. [Paper]

[3] Zhengao Li, Xiaopeng Yuan, Bolin Shen, **Kien Le**, Haohan Wang, Xugui Zhou, Shangqian Gao, Yushun Dong. "**Query-Efficient Domain Knowledge Stealing Against Large Language Models**." *AAAI Conference on Artificial Intelligence (AAAI)*, 2026. [Paper]

Professional Experience

Academic Tutor, Academic Center for Excellence 08/2024 – 07/2026

- Tutor computer science and mathematics one-on-one and in small groups, focusing on problem-solving, algorithmic thinking, and debugging.

Web Development Intern, FPT Software 05/2024 – 08/2024

- Developed full-stack CRUD applications using Java Spring Boot, MySQL, and REST APIs.
- Integrated AWS deployment pipelines and JPA for data persistence.

Projects

GNNFingers Implementation

- Implemented *GNNFingers* (fingerprinting to verify GNN ownership) on PyTorch Geometric; contributed code and docs to the open-source **PyGIP** library.

Waymaker, Knight Hacks IX Devpost

- Offline AI backpack for blind users on an NVIDIA Jetson Orin Nano: YOLO tracking with time-to-collision prediction and ultrasonic fusion for haptic alerts, plus a voice assistant.
- Built the YOLO detection model and quantized it with TensorRT for real-time inference within the Jetson's 8 GB of memory, shared with a vision-language model and speech models.

OpenPotholeMap, ShellHacks 2025 (Waymo Track Winner) GitHub | Devpost

- Crowdsourced real-time pothole detection: phone camera stream, YOLOv8 detection via Roboflow, and GPS-tagged reports on a live map with driver alerts.
- Full stack: React/TypeScript, Node.js/Express, MongoDB, Socket.io, Google Maps API.

Skills

Programming Languages	Python, C/C++, Java, JavaScript, TypeScript, SQL, HTML/CSS
Research Areas	LLM security, adversarial ML, model fingerprinting, federated learning
Machine Learning	PyTorch, PyTorch Geometric, DGL, TensorFlow, scikit-learn, YOLO
Privacy & Edge AI	Homomorphic encryption (TenSEAL), TensorRT, quantization, Jetson
Data & Experiments	NumPy, Pandas, Matplotlib, Weights & Biases
Systems & Web	Linux, Git, GitHub Actions, GDB, React, Node.js, MongoDB, Spring Boot
Languages	English (Proficient), Vietnamese (Native)

Honors, Leadership & Service

2026	Program Committee Member — UrbanAI'26, 4th ACM SIGSPATIAL Urban-AI Workshop
2025	Winner — ShellHacks 2025, Waymo Track
2024	1st Place — FSU ACM Programming Contest (Upper Division)
09/2024 – 12/2024	Honors Colloquium Leader, FSU Honors Program